KUGAYA

GREAT INDIAN AI INTERNSHIP CHALLENGE

AI-Powered Multilingual STEM & Engineering Learning Tutor

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1. THE PROBLEM WE ARE SOLVING

In India, over **85% of engineering students** in Tier-2/3/4 institutions studied in regional vernacular mediums before college. Upon entering STEM programs:

- **Linguistic Intimidation:** Core CS concepts (pointers, deadlock, recursion) are taught strictly in technical English, causing cognitive overload.
- **Hesitation & Memorization:** Students fear asking doubts in English classrooms, resulting in surface memorization without conceptual intuition.
- **Alien Metaphors:** Western textbook examples fail to resonate with everyday Indian student intuition.

2. WHO WILL USE IT (TARGET AUDIENCE)

KUGAYA is built for India's massive engineering student ecosystem:

- **1.5M+ Undergraduate Students:** B.Tech, BE, BCA, and Polytechnic diploma students across all engineering branches.
- **Vernacular-First Thinkers:** Students who reason in regional dialects (Telugu, Hindi, Tamil, Kannada, Malayalam, Bengali) while studying.
- **GATE & Campus Placement Aspirants:** Students needing fast, intuitive mental models paired with strict English exam definitions.

3. THE SOLUTION: VOICE-FIRST 5-CARD PEDAGOGICAL BREAKDOWN

Students ask engineering questions by voice or text. KUGAYA synthesizes a 5-tier multi-modal explanation:

1. Simple Vernacular
Natural code-mixing (e.g. *Tenglish/Hinglish*) to explain core principles intuitively.

2. Cultural Analogy
Relatable Indian life analogies (e.g. Dabba tiffin sets for Recursion, Railway tokens for Mutexes).

3. English Tech Terms
Preserves industry-standard English terminology with clear vernacular context.

4. Exam-Ready Notes
Formal academic answers structured for 8–10 mark university exams.

5. Comprehension Check
Scenario-based check to verify edge cases and conceptual mastery.

4. GNANI AI MODELS & APIS UTILIZED

Gnani Model / API	Endpoint & Technical Role	Impact & Implementation Details
Gnani AI Prisma v2.5 STT	`https://api.vachana.ai/stt/v3` Speech-to-Text Gateway	Enables natural hands-free verbal inquiry in 12 Indian languages. Ingests raw audio with automatic 16kHz mono WAV conversion.
Gnani AI Timbre v2.5 TTS	`https://api.vachana.ai/api/v1/tts/inference` Indic Text-to-Speech	Synthesizes natural mother-tongue audio for each card. Implemented streaming WAV RIFF chunk repair and in-memory caching.
Augmented Reasoning & RAG	Baseten Meta Muse Spark Open + Dense/BM25 Hybrid RAG	Multi-turn pedagogical CoT reasoning grounded in accredited university syllabi (AICTE, JNTU, Anna University) with chapter/page citations.

5. EACH MEMBER'S CONTRIBUTION (KUSHAL GORJI — SOLO BUILDER & ARCHITECT)

Full-Stack Architecture & System Engineering: Built asynchronous FastAPI microservices and React 19 + TypeScript frontend with 8 cyber-vernacular screens, achieving sub-second build times.

Gnani AI End-to-End Integration: Wrote real-time audio pipeline, WebM-to-WAV 16kHz transcoding, WAV header streaming repair, and resilient API fallback mechanisms.

Pedagogical Prompt Pipeline & Hybrid RAG: Engineered the 5-card prompt engine, multi-turn conversational memory, and hybrid dense (256-dim) + sparse (BM25Okapi) textbook retrieval.

Futuristic Neural Orb & Discrete Audio Controls: Created 3D floating hologram orb with cosmic silk ribbons, audio-reactive 7-bar equalizer, and `NeonAudioPlayer` (Pause/Resume/Stop).